

**Carnegie
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University**

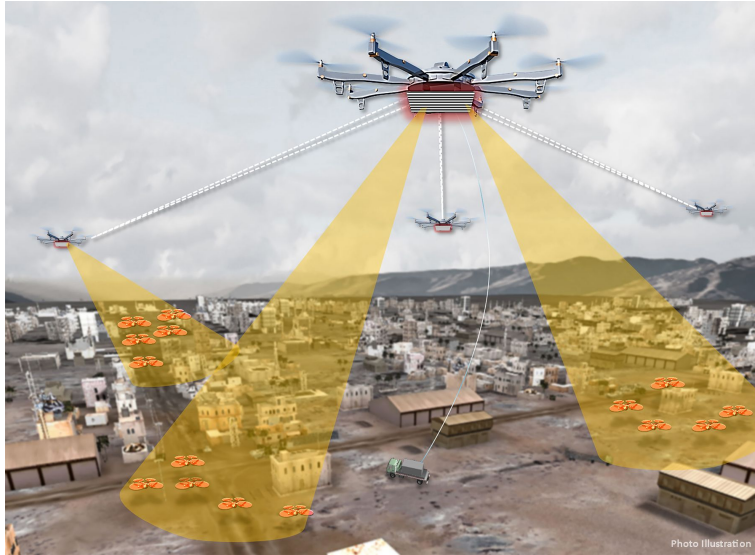
Continuous Sparse Sensing in Ergodic Optimization

Cindy Woo

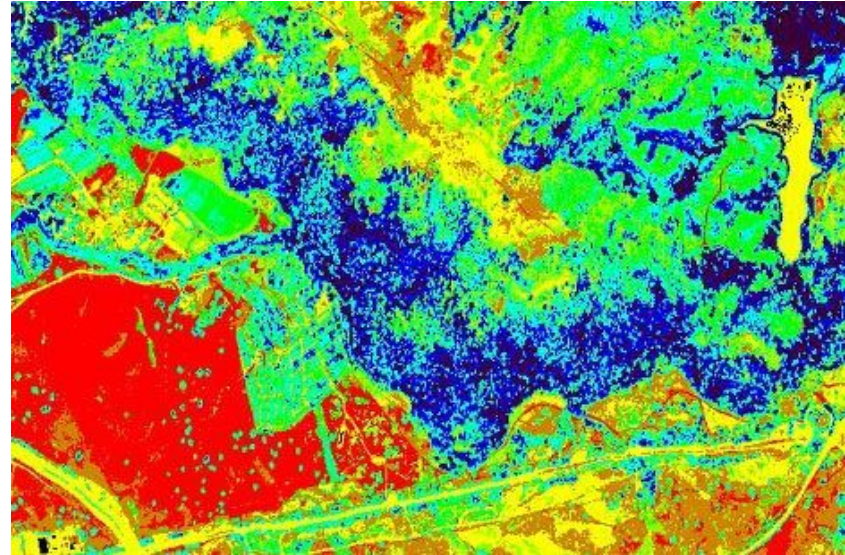
Sebastian Scherer, Howie Choset



Motivation & Core Challenge



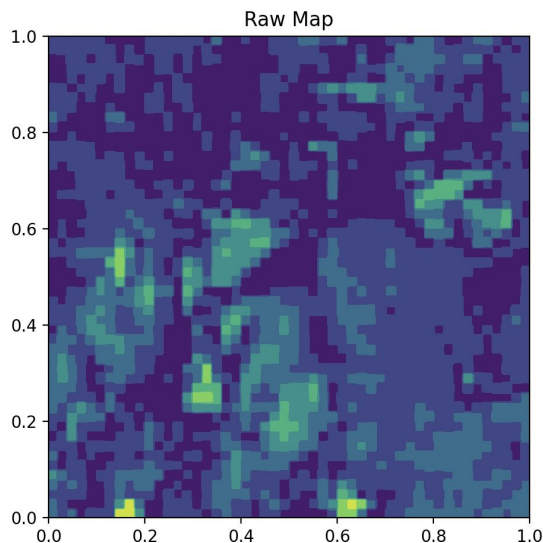
"Always-On" sensing



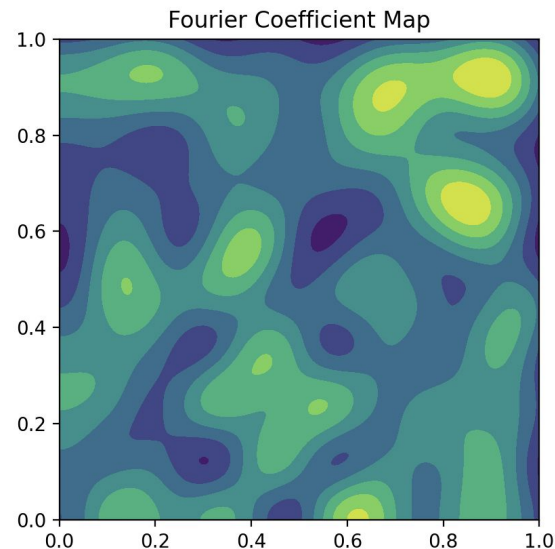
A vast, complex environment needed to be covered

Source: <https://www.livescience.com/56177-darpa-aerial-drag-net-drone-monitoring.html>, <https://www.nv5geospatialsoftware.com/Solutions/Industries/Disaster-Management>

Proposed Solution - Ergodic Exploration & Sparse Sensing

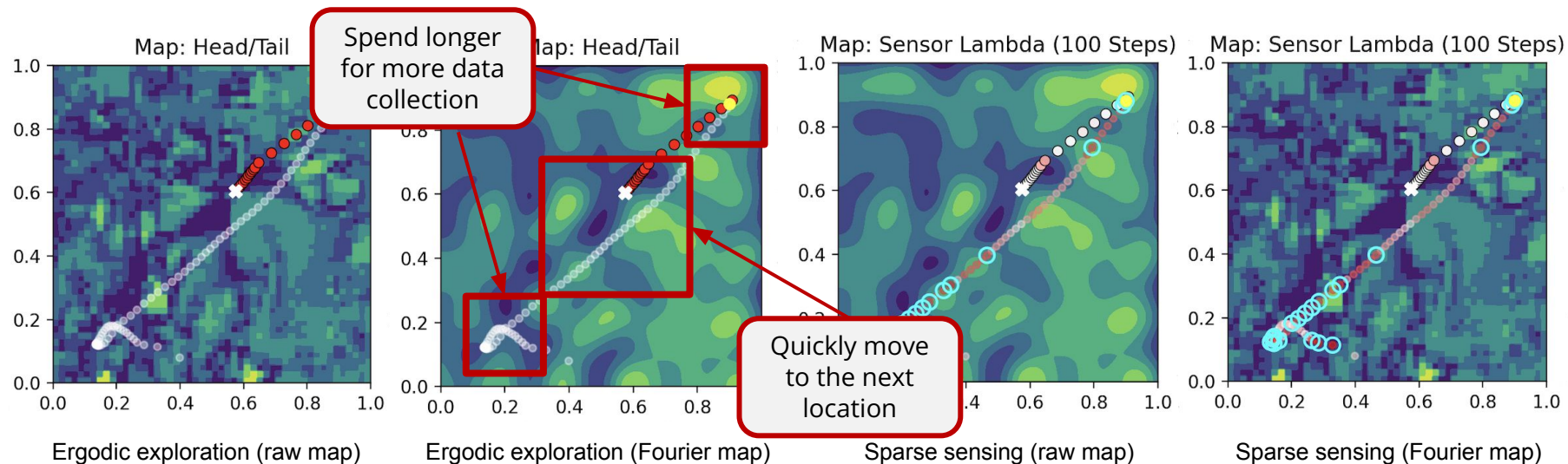


Given raw map input



Fourier coefficient based map

Proposed Solution - Ergodic Exploration & Sparse Sensing



Mathematical Framework - Ergodic Exploration & Sparse Sensing

Fourier Ergodic Loss:

- Minimizes the distance between the trajectory's spatial distribution and the target map's, c_k , spatial distribution, ϕ_k , in the Fourier domain

$$\mathcal{E} = \sum_k \Lambda_k (\phi_k - c_k)^2$$

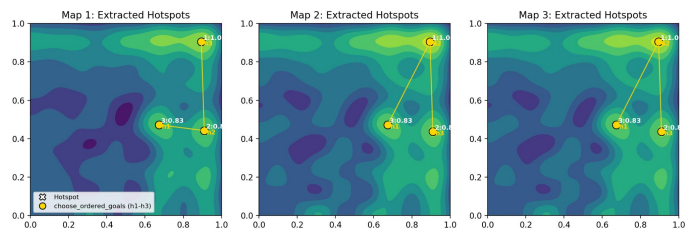
Sparse Sensing Activation:

- Show how the sensor turns on/off continuously using a sigmoid function based on the control input, u

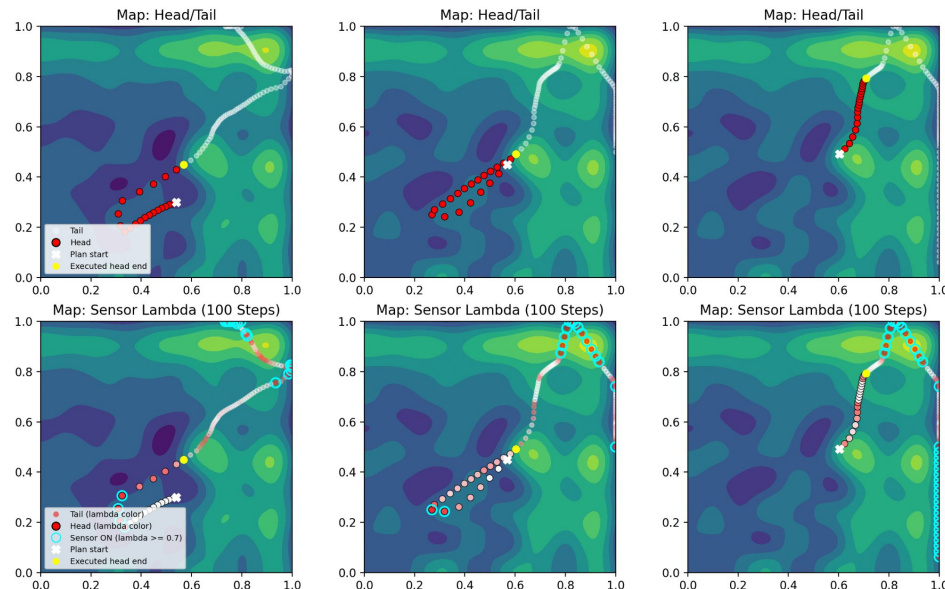
$$\lambda(u) = \frac{1}{1 + e^{-5.0 \cdot u_2}}$$

λ weights the c_k term, meaning the trajectory only "counts" toward coverage when the sensor is active.

Proposed Solution - Receding Horizon Replanning (ITOMP)



3 Hotspots detection per map



Output trajectory per map

Proposed Solution - Receding Horizon Replanning (ITOMP)

Map 1 (New):

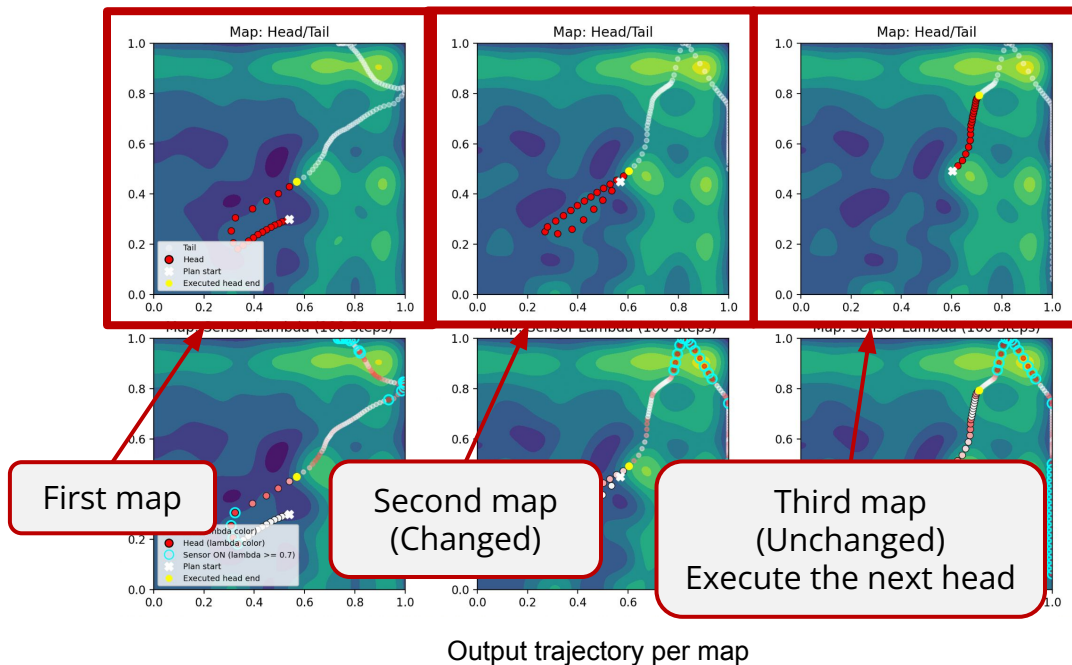
- Optimize head (heavily) and tail (lightly)

Map 2 (Changed):

- Reoptimize head (heavily) and tail (lightly) based on new input

Map 3 (Unchanged):

- Execute next head from previous calculation and maintain remaining tail (w/ 20 additional timesteps)



Mathematical Framework - Receding Horizon Replanning (ITOMP)

1. Map Change Detection

- Calculate the *mean absolute difference* between the new information map and the previous map

$$\Delta = \frac{1}{N} \sum_{i=1}^N |M_{new}^{(i)} - M_{old}^{(i)}|$$

3. Warm Seeding the Receding Horizon

- Replanning: Take the unexecuted portion of the active control plan
- Tail less than the total horizon: Pad the end with the last known control step

$$u_{seed}(t) = \{u_{active}(t + t_{cursor})$$

$$\text{if } t < T_{remaining} \quad u_{active}(T_{total}) \quad \text{if } t \geq T_{remaining}$$

2. Head Quality Evaluation

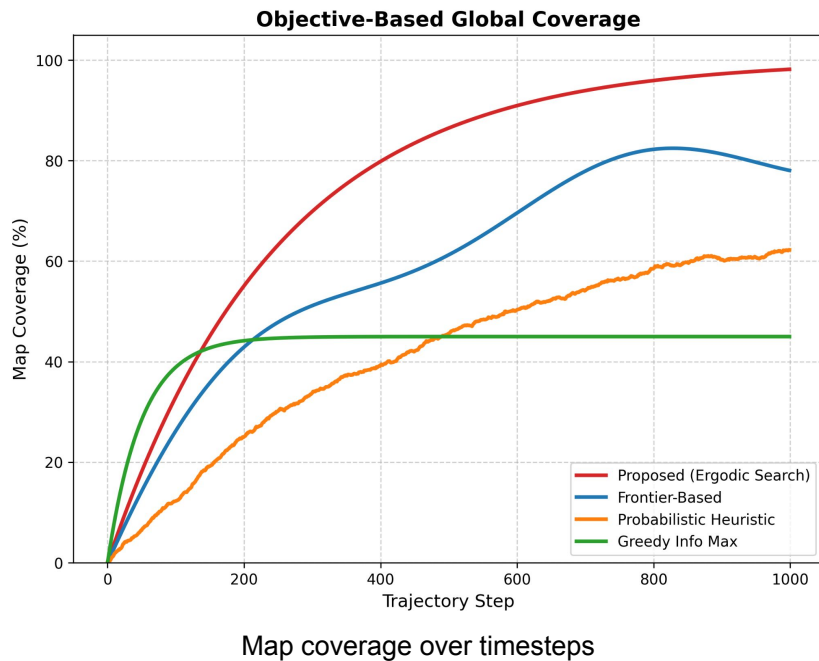
- Unchanged map: Evaluate the current trajectory head's effectiveness toward the first hotspot to determine if a "refinement" replan is needed

$$d_{quality} = \min_{t \in [1, \tau]} \|x(t) - g_1\|_2$$

4. Partial Trajectory Optimization

- Head (Optimized): $u_{head} = u_{seed}[0 : T_{opt}]$
(Requires gradient computation)
- Tail (Frozen): $u_{tail} = u_{seed}[T_{opt} : T]$
(Detached from the computation graph)

Baseline Comparisons



ALGORITHM PERFORMANCE SUMMARY (End of Horizon)

Algorithm	Final Coverage	Final Ergodicity	Avg Time/Cycle
Proposed (Ergodic Search)	98.2%	0.055	0.454s
Frontier-Based	78.0%	0.266	0.764s
Probabilistic Heuristic	63.1%	0.499	0.053s
Greedy Info Max	45.0%	0.550	0.010s

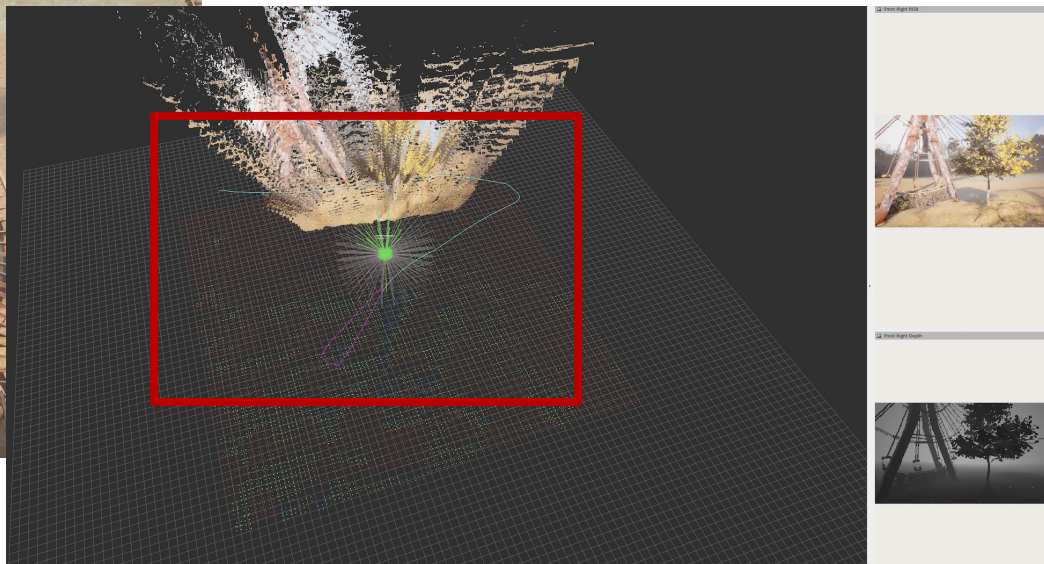
Performance summary comparison

Per map
→ Using ITOMP, Avg time/cycle
significantly decreases

Simulation Architecture & Results

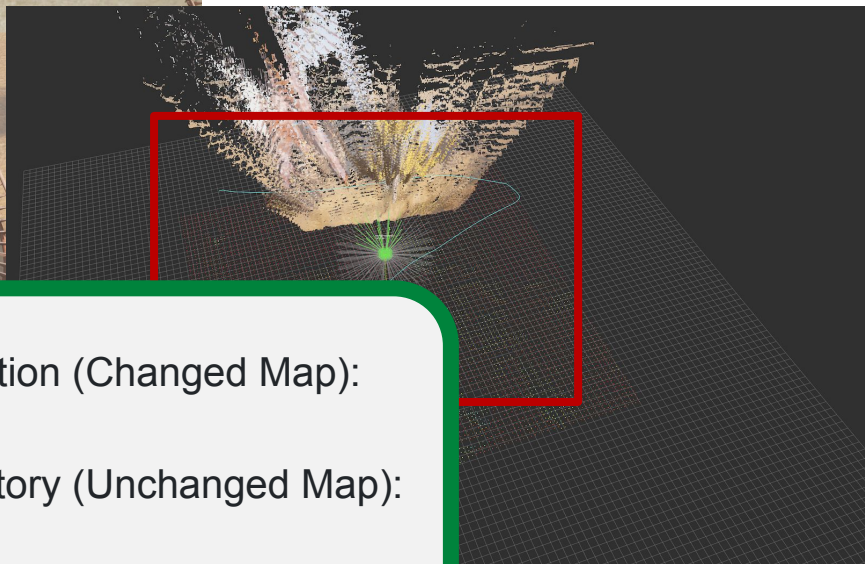


Simulated environment



RVIZ output

Simulation Architecture & Results



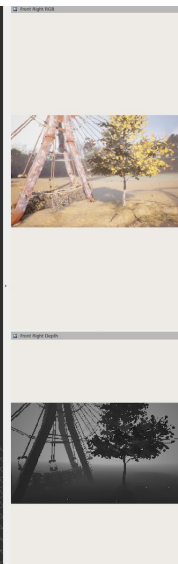
New Trajectory Optimization (Changed Map):

16.2 seconds

Reusing Previous Trajectory (Unchanged Map):

0.02 seconds

RVIZ output



Limitations & Future Work

1. **Spectral truncation** → raw data vs. reconstructed map
 - Blurred or lost data from Fourier domain calculation using a limited number of K coefficients
 - Failed or stuck due to ϕ_k map acting like a low-pass filter
2. **Trajectory smoothing** → Slightly aggressive trajectory
 - Need a smoother, more direct, and dynamically feasible paths for UAVs
3. **Physical drone application** → Modal AI Starling 2 Max



Any Questions?

Thank You!